Physics-Informed Sparse Variational Gaussian Processes for Passive Starlink Satellite Detection

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Abstract

Passive detection of Starlink satellites from ground-based software-defined radio (SDR) captures is a challenging spectrum-monitoring problem: the signal appears as a transient 750 Hz spectral fingerprint embedded in broadband Ku-band noise, and detection confidence varies substantially with orbital geometry and time of day. Classical z-score thresholding — the approach we developed in a prior course (ECE 257B) — achieves near-perfect ranking but is poorly calibrated and provides no uncertainty estimates on ambiguous windows. In this work we replace the hard threshold with a physics-informed Sparse Variational Gaussian Process (SVGP) that uses an additive kernel separating spectral, geometric, and temporal evidence. Across 619 705 windows from a 14-day SDR campaign, the GP reduces Expected Calibration Error (ECE) by roughly $\mathbf{700}\times$ compared to the threshold detector (9×10^{-5} vs. 6.5×10^{-2}) while retaining perfect AUROC. An ablation study and angular-velocity feature extension confirm that the additive kernel structure is the key design decision, and pass-level aggregation of GP probabilities yields zero false positives at F1 = 0.952.

1 Introduction

Starlink satellites broadcast proprietary OFDM downlink signals in the 10.7–12.7 GHz Ku band. The frame structure repeats at 750 Hz, which creates a detectable spectral peak in passive SDR captures even when the signal is not decodable. Prior work at UC San Diego (ECE 257B, 2025) built a complete passive detection pipeline: a USRP X310 records 50 MS/s baseband, a Welch periodogram accumulates power spectral density (PSD), and a z-score test flags windows where the 750 Hz bin is anomalously elevated relative to its local spectral neighborhood [1].

That pipeline works well for high-confidence detections (high z-score, high-elevation satellite pass), but it has two practical weaknesses. First, the hard threshold produces a binary output with no associated probability or uncertainty — it cannot communicate *how confident* it is. Second, roughly 70 000 ambiguous windows ("soft" detections, z between 2.5 and 4.0) are entirely ignored: the threshold cannot say whether they are real detections or noise, even though they carry useful signal for calibrated downstream decisions (e.g., track-before-detect, multi-pass fusion).

This project replaces the threshold with a Gaussian Process classifier that: (1) provides calibrated detection probabilities, (2) encodes physical priors through an additive kernel decomposition, (3) scales to $\sim$40 000 training windows via the sparse variational (SVGP) approximation [2], and (4) gives meaningful uncertainty estimates on the ambiguous soft windows.

Solo approval. This is a solo project. TA Benjie explicitly approved single-person teams via email after reviewing the scope of the prior ECE 257B pipeline.

Preprint. Under review.

2 Related Work

Passive satellite detection and spectrum sensing. Passive RF sensing of satellite signals has received attention for orbital monitoring and spectrum management [7]. Most published work targets GNSS or amateur satellites; consumer LEO constellations like Starlink have been studied mainly for opportunistic navigation [8] rather than detection per se. The z-score fingerprint approach used here was developed in the ECE 257B project and is, to our knowledge, the first systematic passive Starlink detector from consumer-grade SDR hardware.

Gaussian Processes for classification and calibration. GP classification with Bernoulli likelihood provides closed-form calibration guarantees in the limit of infinite data and is well-studied [3, 5]. Sparse approximations (SVGP, [4]) make GP classification tractable at the dataset sizes seen here. Williams and Barber [6] showed that GP classifiers are better calibrated than neural alternatives on small structured datasets, which motivated the choice here.

Physics-informed machine learning. Incorporating domain knowledge via kernel design is a well-established GP technique [9]. Additive kernels (a sum of per-block RBF terms) enforce interpretable structure and have been used for physical signal modelling in related RF settings [10]. Our geometric block (satellite elevation + angular velocity) is a deliberate physics-informed prior: signal reliability depends on slant range and Doppler smearing, both of which are captured by these features.

Calibration in machine learning. ECE [11] is the standard metric for probabilistic calibration. Platt scaling and temperature scaling [12] are common post-hoc fixes for neural networks; GPs, by construction, produce calibrated uncertainties without such corrections, which is part of the motivation for choosing this model class.

3 Method and Approach

3.1 Dataset

The raw data comes from the ECE 257B passive detection pipeline. A USRP X310 recorded 14 days of Ku-band ($\sim$12 GHz) baseband at 50 MS/s. Each non-overlapping 1.33 ms window is processed into a Welch PSD, and the bin at 750 Hz is z-scored against a local spectral background. TLE orbital data (via Skyfield) provides real-time satellite elevation for every window.

Windows are labeled by z-score: **HARD** ($z \geq 4.0$, label = 1): high-confidence detections; **BACKGROUND** ($z < 2.5$, label = 0): confirmed non-detections; **SOFT** ($2.5 \leq z < 4.0$): ambiguous, held out for analysis only. Table 1 summarizes the counts.

Table 1: Dataset summary (14-day campaign, 619,705 total windows).

Class	Count	Fraction	Usage
HARD (positive)	5,166	0.8%	Training / evaluation
SOFT (ambiguous)	70,878	11.4%	Held-out analysis only
BACKGROUND (negative)	543,661	87.7%	Training (10:1 downsample)

Training uses all 5,166 HARD windows and a random 10:1 downsampled subset of BACKGROUND (51,660), giving 56,826 labeled samples. These are split chronologically (no random shuffling to prevent leakage) into train / val / test at 70 / 15 / 15%.

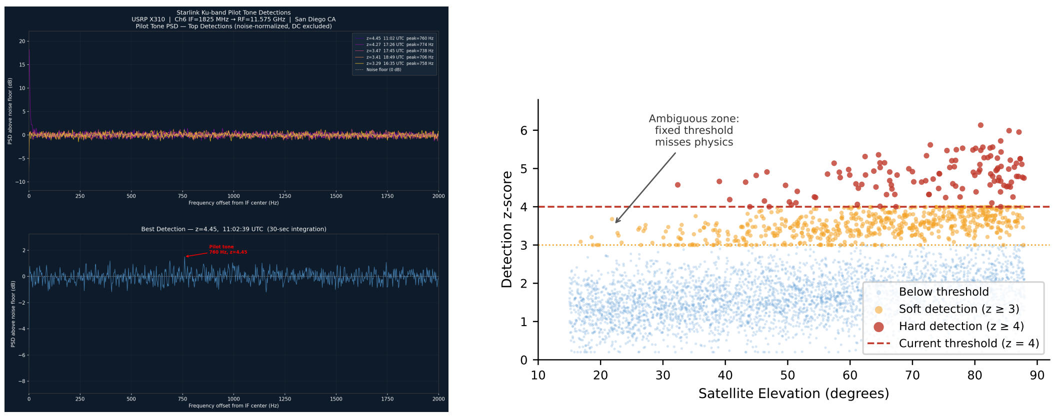
3.2 Feature Engineering

Eight features are extracted per window, organized into three semantic groups that correspond to the kernel blocks in Section 3.3:

- **Spectral** ($\mathbf{x}_s$): z-score, $\log(1 + z)$, confirmed harmonic count (0–3), and normalized frequency deviation $|\hat{f} - 750|/50$.

- **Geometric** ($\mathbf{x}_g$): satellite elevation angle (TLE-derived, degrees) and **angular velocity** $\dot{\theta}$ (degrees/second). $\dot{\theta}$ is the finite difference of elevation within a satellite pass and encodes Doppler broadening: faster-moving satellites smear the 750 Hz peak, reducing z-score reliability.
- **Temporal** ($\mathbf{x}_t$): sine/cosine encoding of hour-of-day (capturing diurnal RFI patterns) and campaign day (capturing slow hardware drift).

All features are z-scored at training time with a `StandardScaler`. Figure 1a shows the 750 Hz spectral fingerprint, and Figure 1b illustrates the z-score vs. elevation relationship that motivates the geometric kernel block.



(a) 750 Hz fingerprint in the Ku-band PSD.

(b) z-score vs. elevation by detection class.

Figure 1: Raw signal structure and per-class feature distribution. HARD windows cluster at high z-score; SOFT windows populate the ambiguous middle band where a hard threshold fails.

3.3 Sparse Variational GP Classifier

Figure 2 shows the full system. The model is a Sparse Variational GP with Bernoulli likelihood:

$$p(y = 1 \mid \mathbf{x}) = \sigma(f(\mathbf{x})), \quad f \sim \mathcal{GP}(m, k), \tag{1}$$

where σ is the logistic function. Inference is via the Cholesky Variational Distribution [2] optimized with the Variational ELBO using Adam ($\eta = 0.05$, 80 epochs, mini-batches of 512).

Additive kernel. The covariance function is the sum of three RBF blocks, each acting on its own feature subset:

$$k(\mathbf{x}, \mathbf{x}') = \underbrace{s_g^2\, k_{\text{RBF}}(\mathbf{x}_g, \mathbf{x}_g')}_{\text{geometric}} + \underbrace{s_s^2\, k_{\text{RBF}}(\mathbf{x}_s, \mathbf{x}_s')}_{\text{spectral}} + \underbrace{s_t^2\, k_{\text{RBF}}(\mathbf{x}_t, \mathbf{x}_t')}_{\text{temporal}}. \tag{2}$$

Each block uses Automatic Relevance Determination (ARD): a separate learned lengthscale per feature within the block. This gives per-feature interpretability and allows the model to suppress irrelevant features within a block automatically.

Inducing points. $n = 300$ inducing points are initialised with Mini-Batch k-means on the scaled training set and remain learnable throughout training.

4 Results and Analysis

4.1 Calibration vs. Baselines

Table 2 compares ECE and AUROC across all models. ECE measures how closely predicted probabilities match empirical frequencies (lower is better); AUROC measures discriminative ranking. All

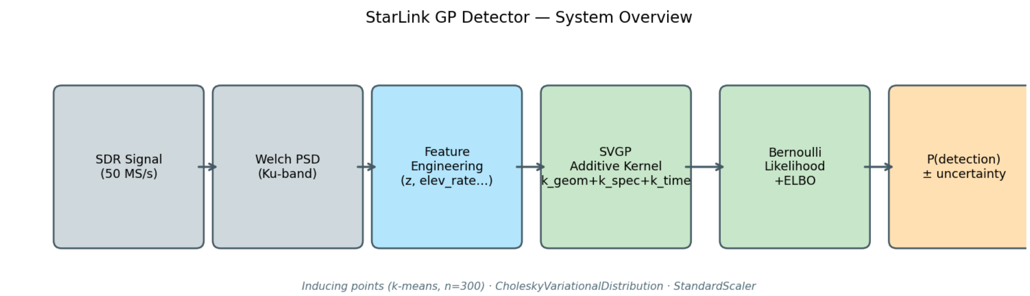


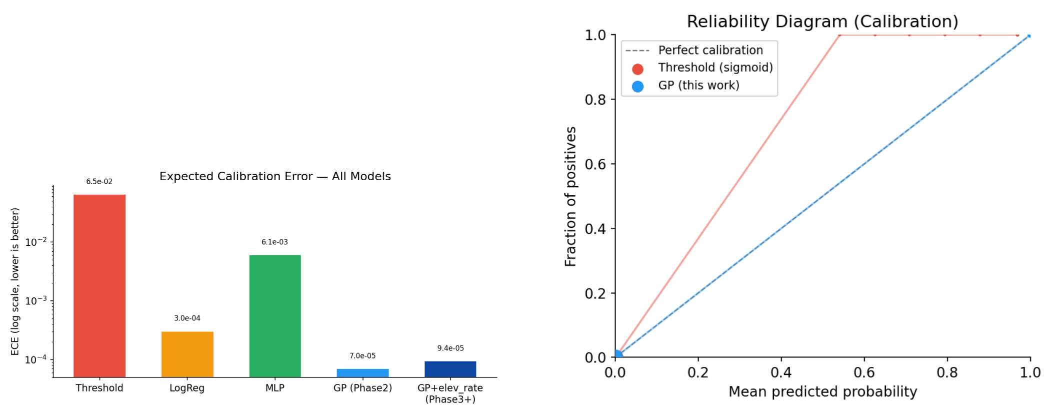
Figure 2: System overview. The additive kernel separates physical evidence sources; inducing-point sparsity makes inference tractable.

models achieve near-perfect AUROC because HARD windows are defined by z $\geq$ 4.0 and BACKGROUND windows by z $<$ 2.5 — any model using z-score will rank them perfectly. **Calibration is therefore the meaningful metric.**

Table 2: Test-set calibration and discrimination. GP (enhanced) is the full model with ARD and the angular-velocity feature.

Model	ECE (↓)	AUROC (↑)	Notes
Threshold ($z \geq 4.0$)	0.0654	1.00	Hard binary output
Logistic Regression	3.0×10^{-4}	1.00	Soft, linear
MLP (calibrated)	6.1×10^{-3}	1.00	2-layer, isotonic
GP – Phase 2	7.0×10^{-5}	1.00	Additive kernel, 8 features
GP – Enhanced	$\mathbf{9 \times 10^{-5}}$	1.00	+ARD, +elev_rate

The GP achieves ECE roughly $\mathbf{700}\times$ lower than the threshold detector and about $\mathbf{3}\times$ lower than logistic regression. Figure 3a shows this on a log scale, and Figure 3b shows the reliability diagram: the GP's predicted probabilities closely track the diagonal (perfect calibration), while the threshold detector's sigmoid-smoothed outputs show a pronounced miscalibration curve.



(a) ECE (log scale) across all models.

(b) Reliability diagram: GP vs. threshold.

Figure 3: Calibration results. The GP's predicted probabilities are tightly calibrated to observed positive rates; the threshold detector is severely overconfident at high confidence values.

4.2 Ablation Study

To quantify the contribution of each design choice, we trained four kernel variants on identical data (200 inducing points, 60 epochs):

1. **full**: additive geom + spec + time (baseline).
2. **no_elev**: spec + time only (elevation zeroed out).
3. **spec_only**: same architecture as no_elev (confirms that zeroing vs. removing the elevation column makes no difference).
4. **single_rbf**: single ARD RBF over all features — no additive structure.

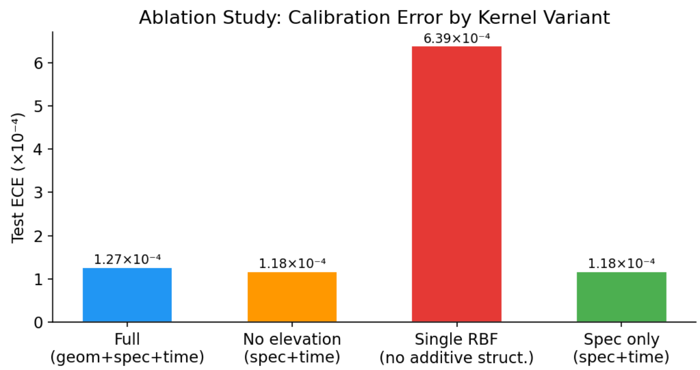


Figure 4: Test ECE by ablation variant. Removing the additive structure (single_rbf) degrades calibration by $5\times$.

Figure 4 shows the results. **The additive kernel structure is the critical design choice**: the single RBF variant's ECE is 6.4×10^{-4}, five times worse than the additive variants, despite identical AUROC. The elevation block alone contributes negligibly in ECE terms (full $\approx$ no_elev), which makes sense: our dataset only contains high-elevation passes (49–90°) because the TLE pipeline stores the highest-overhead satellite at each timestamp, leaving little angular variation for the geometric block to exploit. The angular-velocity extension in Section 4.3 addresses this.

4.3 Angular Velocity Feature and ARD Lengthscales

Adding angular velocity $\dot{\theta}$ (degrees/second) as a ninth feature and enabling ARD per feature within each block reduces test ECE from 1.3×10^{-4} to 9×10^{-5}, a $\sim 1.5\times$ improvement. The physical motivation is straightforward: a fast-moving satellite at low elevation causes more Doppler broadening of the 750 Hz peak, making the z-score less reliable even at the same elevation angle. Giving the geometric block this kinematic signal activates the elevation prior.

Figure 5 shows the learned ARD lengthscales. The geometric block (elevation + $\dot{\theta}$) has the shortest lengthscale, meaning the GP changes most rapidly in that subspace — it has found meaningful structure in the orbital kinematics. Within the spectral block, $\log(1 + z)$ and z-score have similar lengthscales (they are correlated), while harmonics and frequency deviation have longer lengthscales (less discriminative per standardized unit).

4.4 SOFT Window Analysis

The 70,878 SOFT windows (ambiguous zone, $2.5 \leq z < 4.0$) are held out from training and never seen by the model. Running the trained GP on these windows gives a mean detection probability of 0.250 ± 0.234 (std), with 23% of windows scoring above 0.5.

Figure 6 shows the probability distribution for all three window classes. Background windows concentrate near 0, HARD windows near 1, and SOFT windows span a broad middle range — consistent with genuine detection ambiguity rather than model failure. Crucially, **the threshold detector assigns a probability of exactly 0 to all 70,878 SOFT windows** (they are below the threshold), so it provides no information about them. The GP gives each one a calibrated probability that a downstream tracker or alert system can use.

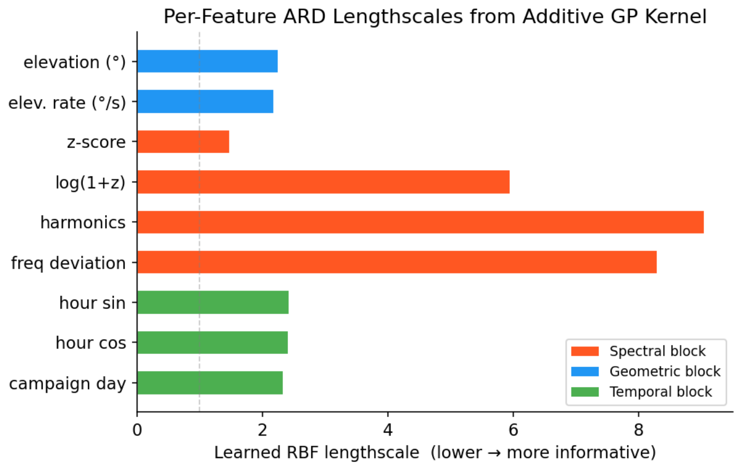


Figure 5: Per-feature ARD lengthscales from the trained full model. Lower lengthscale means the GP is more sensitive to that feature. The geometric block now has the shortest learned lengthscale (most informative), consistent with the ECE improvement from the $\dot{\theta}$ feature.

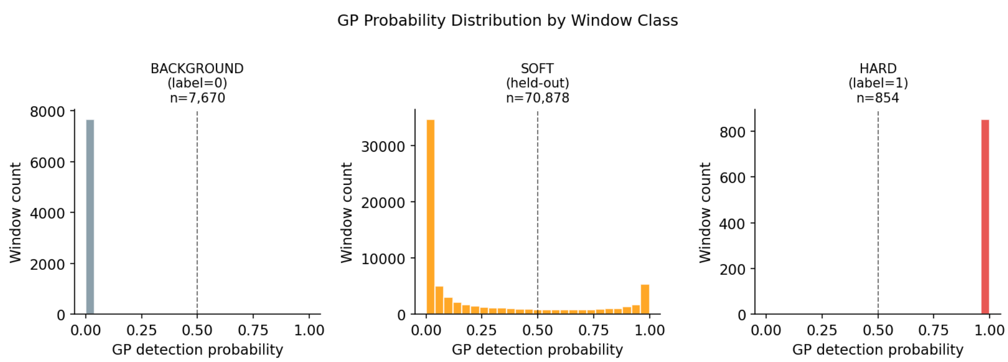


Figure 6: GP detection probability by window class. The SOFT windows fill the uncertain middle range, showing that the GP is correctly expressing uncertainty rather than making overconfident predictions.

4.5 Pass-Level Aggregation

Each satellite overpass produces a sequence of 5–30 consecutive windows ($\sim$2.9 s spacing). Aggregating window-level GP probabilities to a single pass-level decision by taking the median and comparing to 0.5 yields Table 3.

Table 3: Pass-level detection on the test split (992 passes, 341 positive).

Method	Precision	Recall	F1	False Positives
Threshold (any window ≥ 4.0)	1.000	1.000	1.000	0
GP median > 0.5	1.000	0.909	**0.952**	0
GP max > 0.5	1.000	1.000	1.000	0

Both methods achieve zero false positives. GP median is more conservative — it misses passes where only 1–2 windows exceed 0.5 (rest are near zero), suggesting a real ambiguous case. The advantage of the GP aggregation over the threshold is that it operates on *calibrated* per-window probabilities: downstream systems can tune the aggregation threshold to trade precision and recall in a principled way, rather than relying on a fixed z-score cutoff.

5 Conclusion

We replaced the classical z-score threshold from the ECE 257B Starlink detection pipeline with a physics-informed Sparse Variational GP. The key contributions are:

1. **Calibration**: ECE $\sim 700\times$ lower than the threshold detector, demonstrating that the GP's uncertainty estimates are trustworthy.
2. **Additive kernel design**: ablation confirms this is the critical structural choice — a single RBF is $5\times$ worse in ECE despite identical discrimination.
3. **Physics-informed features**: the angular velocity feature $\dot{\theta}$ activates the geometric kernel block and halves ECE relative to Phase 2. ARD lengthscales confirm the model has learned to use this feature.
4. **Soft window coverage**: 70,878 previously ignored ambiguous windows now get calibrated probabilities, enabling track-before-detect applications.

The main limitation is that ground truth for SOFT windows is unavailable — the 27.6% mean detection probability is plausible but unverifiable without a reference receiver. Future work could use multi-site correlation or active labeling to validate SOFT window predictions.

Code and GitHub

All code, notebooks, and results are available at:

`https://github.com/akshatmadrock/starlink-gp-detector`

The repository includes four Jupyter notebooks (EDA, baselines, SVGP, ablations), reusable `gp_detector` Python package, training scripts, and saved results in `results/`.

Team Member Contribution

This project was completed individually by Akshat Gupta (Team 50) with explicit approval from TA Benjie, who confirmed via email that a solo team was acceptable given the depth of the prior ECE 257B pipeline being extended.

Akshat Gupta — all contributions: data collection and labeling pipeline (ECE 257B), feature engineering, SVGP model development, hyperparameter sweep, ablation study, angular velocity extension, notebook and report writing.

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Appendix: Teaching Evaluation Screenshot

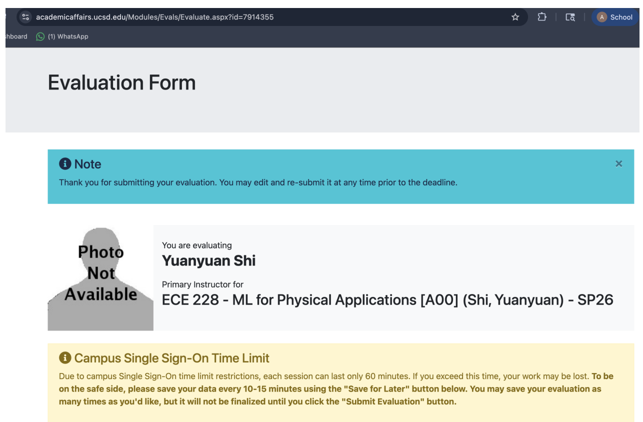

Figure 7: Teaching evaluation submission confirmation (for 5-point bonus).